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TEMPERATURE-RESPONSIVE NEEDLES FOR IMPROVED HEMODIALYSIS PATIENT CARE

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ABSTRACT

In the United States, hemodialysis treatments commonly use medical-grade stainless needles. However, their rigidity often causes patient discomfort and vascular injury, adversely affecting treatment compliance. This study presents a novel needle fabricated from a thermoplastic polymer blend of Poly(methyl methacrylate) (PMMA), Polypropylene carbonate (PPC), and Polyvinyl acetate (PVAc), designed to transition from a rigid to flexible state at body temperature. Tensile tests of our polymer blend produced a tensile modulus of 1.73 GPa at 24°C and 580 MPa at 37°C, corroborating that the needle remains sufficiently rigid during insertion and flexible to conform to the vasculature. This innovative needle design, with its dual-phase functionality, allows for vascular access and is seamlessly adoptable by practitioners accustomed to medical-grade stainless steel needles. Its glass transition behavior directly improves patient comfort, thereby improving patient adherence to regular hemodialysis treatments.

Keywords: Hemodialysis, Arteriovenous Fistula, AVF Access Devices, Thermoplastic Elastomers, Patient Comfort, Conforming, Stenosis, Kidney dysfunction, Blood, Needle, Cannulation, Catheter tubes, Veins, Arteries, Hematology

1. INTRODUCTION

Chronic Kidney Disease (CKD) is a significant health challenge that impairs kidney function and disrupts blood filtration, affecting 38 percent of individuals over the age of 65 in the United States [1]. As CKD advances to End-Stage Kidney Disease (ESKD), hemodialysis becomes essential and provides life-sustaining treatment for approximately half a million Americans [2]. This procedure relies on an arteriovenous fistula (AVF) to facilitate the extracorporeal filtration of blood. Despite ongoing advancements, challenges in patient comfort and treatment adherence persist [3, 4].

Hemodialysis treatments conventionally use medical-grade stainless steel needles due to their low cost, established use in clinical practice, and rigidity, favored for enhanced control during insertion [5]. However, the rigidity of these needles can cause

significant discomfort and vascular injury because of the mismatch in stiffness between the needle and vascular tissues. This mismatch increases friction and stress at the puncture site, heightening the risk of complications such as infiltrations, scarring, and stenosis. These issues can severely impact patient compliance, as discomfort and fear of pain become barriers to adherence to the prescribed hemodialysis regimen [3, 4].

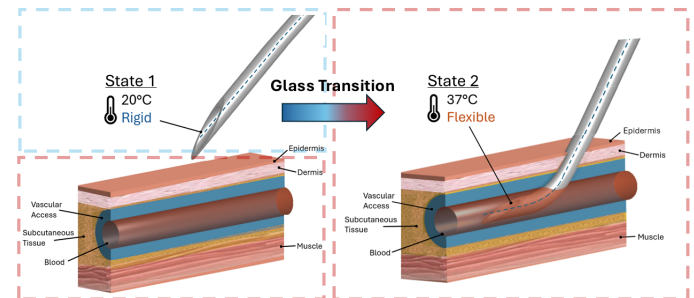


FIGURE 1: FLEXIBILITY TRANSITION DUE TO BODY TEMPERATURE

Hemodialysis sessions in the United States typically last three to five hours and occur three times a week—or indefinitely for ESKD patients. Even minor patient movements during these lengthy procedures can lead to needle agitation, contributing to vascular complications and further discomfort. Once the needle is inserted, patients typically experience no pain unless they move, causing the steel needle to press against the wall of the vein, which can be painful. In contrast, flexible intravascular catheters, like IVs, do not cause discomfort upon movement [6]. Approximately 20% of patients experience severe pain during needle insertion, even with the use of analgesics [7], where pain is quantified using the visual analogue scale (VAS) scale [8]. When fistula infiltration causes pain and discomfort, it can lead to an elevated risk of needle dislodgement, followed by vascular infiltration. This sequence of events may result in swelling, endothelial damage, and significant complications like stenosis and thrombosis[9]. As a result, the overall pain and potential complications surrounding this procedure drive about 30% of patients to skip or shorten their sessions [3], increasing their risk of hospitalization and mortality

by as much as 30% for each missed session [10].

To address these challenges, flexible catheters provide an alternative to medical-grade stainless steel needles. Nevertheless, their use requires some kind of guide device, as seen in the Seldinger technique for precise placement—a procedure involving an introducer, sheath, and dilator to ensure smooth insertion and maintain directional control within the vascular access site. Although this technique reduces vascular trauma, it requires additional training and expertise, thus limiting its accessibility [11]. Consequently, the adoption of catheters has been limited, particularly in settings with inadequate training infrastructure.

As a result, we propose a novel hemodialysis needle design that merges the benefits of both conventional needles and catheters. The needle is composed of a blend of thermoplastic polymers, including Poly(methyl methacrylate) (PMMA), Polypropylene carbonate (PPC), and Polyvinyl acetate (PVAc). The innovative feature is its ability to transition from a rigid state during insertion, which ensures precise directional control, to a flexible state at body temperature, allowing it to conform to vasculature. Tensile testing confirms that the modulus of the polymer blend decreases significantly from 1.73 GPa at 24°C to 580 MPa at 37°C—demonstrating its ability to adapt flexibly within the body while retaining the necessary rigidity during insertion.

2. MATERIALS AND METHODS

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2} \quad (1)$$

For this study, $K=1$ was used to calculate the minimum tensile modulus needed to prevent needle buckling. Since the maximum insertion force is not immediately reached upon insertion, the actual modulus required may be lower due to tissue reinforcement, especially with a low-friction coating. For a 15-gauge hemodialysis needle (1.829 mm OD, 1.372 mm ID, 1-inch length), the calculated minimum tensile modulus needed is approximately 0.5 GPa for an insertion force of 3 N [12]. Conversely, with a tensile modulus of 1.5 GPa, the critical buckling load is 8.6 N. In hemodialysis treatments, 15-gauge hemodialysis needles are typically used when the fistula has matured.

2.1 Polymer Blend Preparation

The study employed a polymer blend with a 70/30 weight ratio of PMMA to PPC, and 5 phr of PVAc. The PMMA, with an average molecular weight of approximately 350,000, was sourced from Millipore Sigma. The PPC (QPAC40), ranging from 100,000 to 300,000 in molecular weight, was obtained from Empower Materials. The PVAc, with a molecular weight of 184,100, was provided by Scientific Polymer Products Inc. PMMA has a tensile modulus of 2.9 GPa, whereas PPC is 34-41 MPa and serves as the plasticizer.

This blend underwent melt-mixing in the Xplore HT 15 Micro-compounder at 210°C and 40 rpm for 5 minutes per batch. The material was then formed into ASTM D638 Type V tensile bars using the Xplore plunger injection molding machine at 190-200°C.

2.2 Mechanical Testing Protocol

After conditioning the bars for 40 hours, three samples were tested at room temperature, approximately 23°C, and at body temperature, around 37°C, using an oven. The tensile testing was performed on an Instron machine equipped with a mechanical extensometer, at a strain rate of 10 mm/min. Maintaining the 37°C temperature during testing was challenging.

2.3 Characterization of PVAc Effects on Polymer Blend

Polyvinyl acetate (PVAc) significantly enhances the mechanical properties of the PMMA/PPC blend by promoting miscibility. This results in a transition from a sea-island to a co-continuous phase structure, which drastically multiplies both the tensile strength and elastic modulus of the blend. The addition of PVAc facilitates this enhancement by reducing interfacial tension and increasing compatibility between PPC and PMMA phases. PVAc shifts the glass transition temperature (T_g) of the blend higher than pure PPC which is around 23-24°C. This is because the glass phase temperature of PMMA is higher than pure PPC [13].

2.4 Hemocompatibility

Polymethyl methacrylate (PMMA) is recognized for its hemocompatibility, making it suitable for blood-contacting applications [14]. Conversely, polypropylene carbonate (PPC) has poor hemocompatibility due to its hydrophobic nature, which can lead to thrombus formation [15]. Surface modifications, such as layer-by-layer (LbL) assembly with anticoagulant molecules, are used to improve the hemocompatibility of PPC [16]. Studies suggest that Polyvinyl acetate (PVAc) can enhance blood compatibility in some polymers [13]. Further studies into the polymer blend at specific concentrations can clarify if the blend is sufficiently hemocompatible. A hemocompatible coating with good adhesion to the polymer blend surface can be an alternative [16].

3. RESULTS AND DISCUSSION

The tensile testing of the polymer composites was conducted using the Instron 4466 Universal Testing Machine, as detailed in Section 2.2. This testing provided critical tensile characterization data, enabling the calculation of Young's Modulus in both the rigid and flexible states. Needle deflection was analyzed to understand the distribution of von Mises stress during insertion, showing that the stresses are more efficiently managed in the glass phase state.

3.1 Design Scope

At 23°C, the polymer blend exhibits a median tensile modulus of 1.76 GPa. Upon reaching the glass transition temperature, this modulus drops to 580 MPa, demonstrating the plasticizing effect of PPC, which facilitates the transition to a flowable state at elevated temperatures. Pure PPC loses strength above 35°C, becoming essentially non-resistant to deformation [17].

The tensile stress-strain graph in Figure 2 demonstrates the behavior of a polymer blend designed for a flexible needle at 23°C and 37°C. At 23°C, the blend shows higher strength and stiffness,

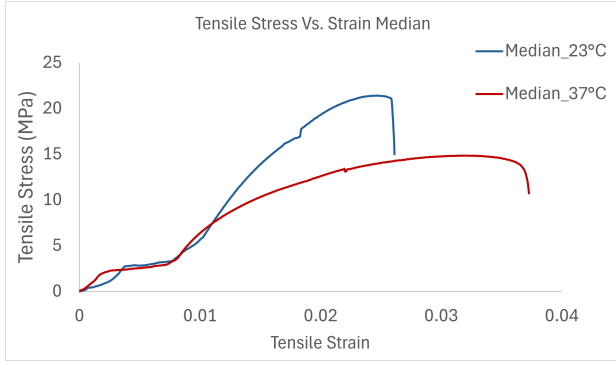


FIGURE 2: MEDIAN TENSILE STRESS AND STRAIN GRAPH OF SAMPLES AT DIFFERENT TEMPERATURES

with a peak stress above 20 MPa before failure, indicating that it is more rigid at room temperature. At 37°C, which approximates body temperature, the blend exhibits lower strength with a peak stress around 15 MPa, and greater ductility, suggesting it softens and becomes more flexible.

TABLE 1: INSTRON TEST DATA FOR 70:30 (PMMA:PPC) WITH 5 PHR PVAC BLEND

Temperature (°C)	Tensile Strain at Max Force [%]	Force at Break [kN]	Tensile Stress at Break [MPa]
23	2.47	0.27	24.17
	1.80	0.19	16.79
	2.38	0.20	17.90
37	2.11	0.18	15.52
	3.33	0.16	14.48
	3.05	0.11	10.38

At room temperature (23°C), the needle material demonstrates moderate ductility with an average tensile strain of 2.22% at maximum force. It also exhibits relatively high mechanical strength, with a breaking force of 0.22 kN and a tensile stress at break of 19.62 MPa.

However, as the temperature increases to 37°C, which is close to body temperature, the material becomes more ductile, allowing for a greater tensile strain of 2.83% before breaking. This increased flexibility is accompanied by a reduction in strength, with the force at break dropping to 0.15 kN and tensile stress at break reducing to 13.46 MPa.

These findings suggest that at body temperature, the needle material will be more flexible, which could enhance patient comfort by allowing the needle to conform better to the tissue. However, this comes with a trade-off in strength, meaning the needle might be more prone to deformation under stress, which could lead to kinking.

Li et al.[13] document a significant decrease in storage modulus by several orders of magnitude for a 70/30 PPC/PMMA blend with 5 phr PVAc, recording a tensile modulus of 317 MPa at ambient conditions. For catheters made of polyurethane, the typical elastic modulus is approximately 55 MPa [18], though these catheters typically feature much thicker walls. In contrast, our design focuses on a 15-gauge needle with significantly thinner

walls.

In this study, the tensile modulus at ambient conditions, measured at 1.76 GPa, greatly exceeds the buckling resistance threshold of approximately 0.5 GPa. This shows the ample scope for modulus reduction at body temperature while adhering to buckling constraints.

The design objective is to lower Young's modulus as much as feasible at body temperature, aiming to improve flexibility while ensuring that the flow rate of the needle does not decrease by more than half due to kinking, an issue identified in flexible catheters by the FDA [19].

Both this study and Li et al.[13] utilized a strain rate of 10 mm/min, which may overestimate the tensile modulus due to insufficient time for polymer deformation at this relatively high rate. A lower strain rate should be used to match the conditions of hemodialysis needle insertion. The tensile bars were preheated to 37°C using an oven; however, sustaining this precise temperature throughout the Instron testing phase was problematic, which might result in an overestimated tensile modulus at the intended body temperature condition. While the tensile bars underwent conditioning, using a larger sample size and implementing a more rigorous post-processing protocol could enhance mechanical reliability and minimize defects in the tensile bars and in the needle itself.

3.2 Computational Simulations

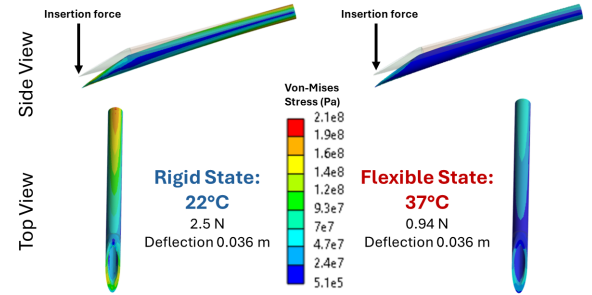


FIGURE 3: STRESS CONTOUR PLOTS OF POLYMER COMPOSITE NEEDLE OF BOTH STATES USING ANSYS

ANSYS Workbench was used to conduct the deflection and buckling finite element analysis (FEA) simulations for the polymer blend for the rigid and flexible state. Ambient and body temperatures for all simulations were set to 22°C and 37°C, respectively. For the needle geometry and design, Wang et al. [20] identify 15 degrees as the optimal zeta angle for IV needle tips, shown in Figure 1. As for the gauge of the needle, this study selected a 15 gauge (15G) needle based on its ability to meet the buckling constraints critical in dialysis treatments for withstanding axial loads without bending or collapsing [21]. Input values of the moduli values provided by the tensile testing and the insertion force required to puncture skin [12] were used for Figure 3. This figure compares the force needed to deflect the needle by the same displacement at different temperatures under the loading configuration, displaying stress contour plots with a side and top view. Corroborating the buckling insights from Equation 1, a

Young's Modulus of 1.5 GPa could withstand up to 8.6N for a 15G needle and the usual penetration depth of 6-10 mm [22].

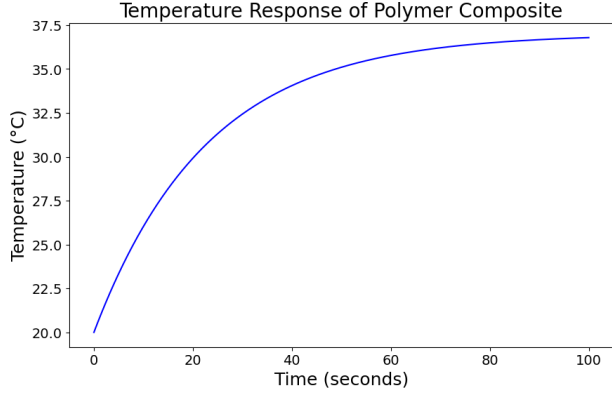


FIGURE 4: THEORETICAL THERMAL RESPONSE OF 15-GAUGE NEEDLE AT HUMAN BODY TEMPERATURE (37°C)

3.3 Time to Reach Glass Phase

To find the thermal time constant, a conservative estimate is taken by choosing the smaller density and specific heat capacity values from each of the three components of our composite.

TABLE 2: PROPERTIES OF DIFFERENT MATERIALS

Materials	ρ [g/mL]	c_p [J/g]
PVAc	1.17	1.63
PMMA	1.17	1.48
PPC	1.26	1.70

The thermal time constant is about 20 seconds, according to Equation 2. The highest applicable forced convective heat transfer coefficient was chosen for the fastest response time, at about $30 \frac{W}{m^2 \cdot K}$.

$$\tau = \frac{\rho V c_p}{h A} \quad (2)$$

The τ constant can model how long it takes for the polymer mix to reach the glass phase temperature. From Equation 3, it takes 6 seconds for the needle to reach glass phase temperature at 24 °C for a polymer composite [17], sufficient for needle insertion and adjustments before being clamped down. Needle insertion typically lasts for 1-2 seconds.

$$T(t) = T_{\infty} + (T_0 - T_{\infty}) \cdot e^{-\frac{t}{\tau}} \quad (3)$$

The exponential heating equation can generate a plot of the overall needle in Figure 4.

4. EXPERIMENTAL PROTOTYPE

Polydimethylsiloxane (PDMS) phantoms were designed to assess the insertion dynamics and vascular interaction of hemodialysis needles under various conditions. By adjusting the bend radius of the phantom channel, the setup could simulate different anatomical scenarios, allowing for the testing of needle adaptation to varying vascular curvatures. This method was

used to determine how well the prototype needle performs during insertion at different angles and radii, which is critical for evaluating the risk of needlestick injuries and other vascular damage associated with traditional rigid stainless steel needles.

The prototype needle manufactured for concept testing incorporated the same 70:30 PMMA:PPC mixture with 5 phr PVAc. The polymers were blended in acetone, which was later evaporated in an oven at 200°C. This was then injection molded using a Morgan Press into an aluminium injection mold, manufactured using CNC machining techniques. This mimicked the geometry of a 15 gauge needle, with an inner rod to capture the needle geometry. Given that acetone does not contribute to the stiffness of the composite, it was necessary to normalize the cross-sectional area to reflect only the remaining polymer content. To simulate the conditions upon insertion into the body, the polymer composite was immersed in a thermal bath at 37°C.

4.1 Vein Phantom

To test the prototype needles and the proof of concept of the benefit of a rigid to flexible state, a vein phantom was cast from a polydimethylsiloxane (PDMS) mixture, chosen for its elastomeric properties that closely mimic the stiffness of human fistulas. The phantom was fabricated from a Sylgard 184 kit, employing a 10:2 weight ratio of PDMS to curing agent for the desired 1.5MPa stiffness of a human fistula [23, 24]. This mixture was poured into a 3D printed PLA mold with an internal rod. It was allowed to cure at room temperature for 24 hours. After curing, the PDMS was removed from the mold, and the rod was extracted, leaving a channel that simulated the AVF's lumen. Figure 5a shows a medical-grade stainless steel needle puncturing through the opposite wall, while Figure 5b depicts the needle being dislodged during different vein movements. Figure 5c shows the needle inserted into a straightened vein phantom. In contrast, Figures 6a and 6b demonstrate the softened needle composite adapting to the same vein movements without puncturing the opposite wall or becoming dislodged, respectively. Figure 6c illustrates the needle conforming to a straightened vein phantom.

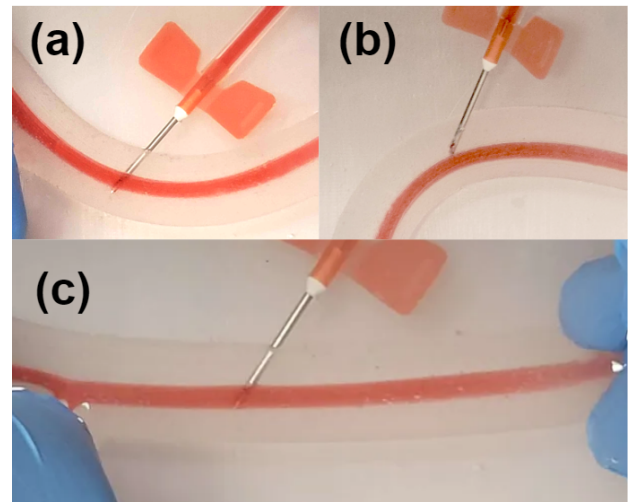


FIGURE 5: MEDICAL-GRADE STAINLESS STEEL NEEDLES INSERTED INTO VEIN PHANTOMS

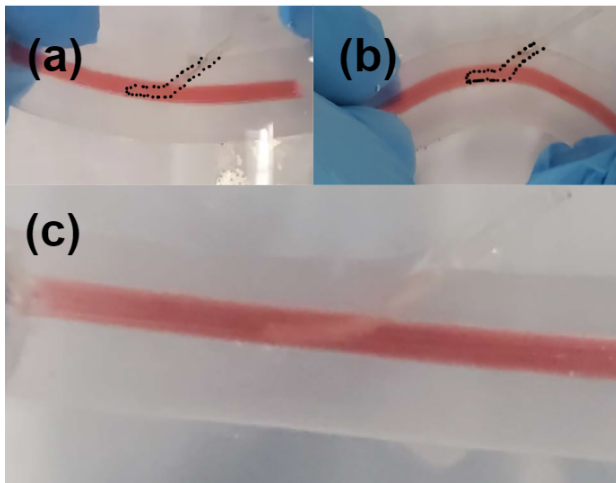


FIGURE 6: SOFTENED POLYMER COMPOSITE NEEDLES ARE INSERTED INTO VEIN PHANTOMS

5. CONCLUSION

This study demonstrates a broad design scope for optimizing this novel needle polymer blend that withstands buckling upon insertion while strategically reducing tensile modulus at body temperature, effectively mitigating vascular damage without risking needle kinking. By addressing the mechanical compatibility between the needle and vascular tissues, this design can enhance patient comfort, adherence to treatment, and overall hemodialysis outcomes, paving the way for significant advancements in dialysis care.

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